



Angular CLI and Project Setup

Persistent Interactive | Persistent University



Key Learning Points

- Install Angular CLI
- Set up a new Angular Project
- Understand project structure
- Concept of Components
- Concept of Modules

Angular CLI

- It is a **command line interface** for working with Angular projects.
- It is a tool available on **NPM** and needs to be installed using the NPM commands.
- Enables to create a **new Angular project** with all dependencies (**node_modules**).
- Helps to **create modules, components, services, pipes, directives and routes** with very simple commands.
- Creates the **basic file structure** with appropriate extensions.
- Generates the **boilerplate code** for every file type along with default import statements.
- Also enables to **build and run** an Angular project with live-reloading of browser i.e any changes in code will automatically get reflected in the browser.

Installation

- Ensure Node.js is installed on the system.
- NPM (Node Package Manager) comes along with Node installation.
- NPM enables to download packages from <https://www.npmjs.com/>
- NPM has a lot of commands to install/update/delete/check version of modules on <https://www.npmjs.com/>
- Run the below command to install Angular CLI

```
npm install -g @angular/cli
```

- **The flag `-g` indicates it will be installed as a command line tool**

Ivy Compiler

- Version 9 moves all applications to use the Ivy compiler and runtime by default.
- First, update to the latest version of 8
 - `ng update @angular/cli@8 @angular/core@8`
- Then, update to 9
 - `ng update @angular/cli @angular/core`

Ivy Compiler in Angular 9.0

- Smaller bundle sizes
- Faster testing
- Better debugging
- Improved CSS class and style binding
- Improved type checking
- Improved build errors
- Improved build times, enabling AOT on by default

Create a new Angular project

- Run the below command to create a new Angular project

```
ng new <PROJECT-NAME>
```

- The below will create a new project with name 'demoProject'.

```
ng new demoProject
```

- With this command, it will internally also fetch all the module dependencies i.e Angular modules(@angular/core,@angular/http etc) as well as some popularly used external modules (rxjs).
- All project dependencies are listed in **package.json** file.

Run the Angular Project

- Open the Project in VSCode Editor.
- Open the Integrated terminal in VSCode.
- Install Angular CLI in VSCode as well.

```
npm install -g @angular/cli
```

- This runs a development server (written in Node.js) and serves the Angular application. Also runs **tsc** in watch mode.
- Default URL is <http://localhost:4200>
- URL and port can be configured with command :

```
ng serve --host 0.0.0.0 --port 4201
```


From Node.js Command Prompt

- Browse to the generated project folder.

```
cd demoProject
```

- Serve the project with below command

```
ng serve
```

Angular App project structure

- The important files
 1. **angular.json:** Configures how the Angular modules are to be loaded.
 2. **package.json:** List of module dependencies
 3. **index.html:** Shell page for the Single page App
 4. **main.ts:** Bootstrapping of Root module happens
 5. **'src/app' folder:** Contains all source files

App folder contents

- **'src/app' folder** contains all source files
- app.module.ts
- app-routing.module.ts
- app.component.ts
- app.component.html
- app.component.css

Angular Application

AppModule

AppComponent

What are Components?

- **Components** are the basic building blocks of an **Angular** application.
- A **component** defines a patch of screen.
- Examples include:
 - Navigation Menu
 - Login Panel
 - Products List (name, price, quantity)
 - Shopping Cart
- Component files are named with **.component** suffix
- Every Component has its own Model, View and Controllers.

Add a Component to page

- Every HTML element has a tag associated with it.
- `<div></div>`, `<form></form>`
- Every component too has a selector.
- To add a component to page use the associated selector.
- **The index.html page has the selector `<app-root></app-root>`**

Modules contain Components

- Components are organized with the help of modules.
- Every Module can consist of multiple components and other constructs like directives, pipes and services.

Module

Component1

Component2

Component3

Component4

- Module files are named with **.module** suffix.

Angular Application

AppModule

AppComponent

Generate Components

- Command to generate a component

```
ng generate component login-component
```

```
ng g component login-component
```

- Creates a folder with component name containing the ts, css and html files.
- angular-cli will add reference to the newly created component automatically in the **app.module.ts**. **That means it adds an entry to the declarations array.**

Create component in feature module

- If component to be created in a feature module, use below command.

```
ng g component loginmodule/login-component
```

- **With this command it will add reference of component to the corresponding feature module file.**

Generate other Angular constructs

Scaffold	Usage
Component	ng g component my-new-component
Directive	ng g directive my-new-directive
Pipe	ng g pipe my-new-pipe
Service	ng g service my-new-service
Class	ng g class my-new-class
Guard	ng g guard my-new-guard
Interface	ng g interface my-new-interface
Enum	ng g enum my-new-enum
Module	ng g module my-module

Debugging with Angular 9 Ivy

- A big improvement introduced in Angular 9 and Angular Ivy is the debugging experience.
- The compiler now can detect more errors and throw them in a more “readable” fashion.
- We can activate template checking in tsconfig.json with below property.
- "angularCompilerOptions": {
- "fullTemplateTypeCheck": true,
- "strictInjectionParameters": true,
- **"strictTemplates": true**
- }

Bundle Size

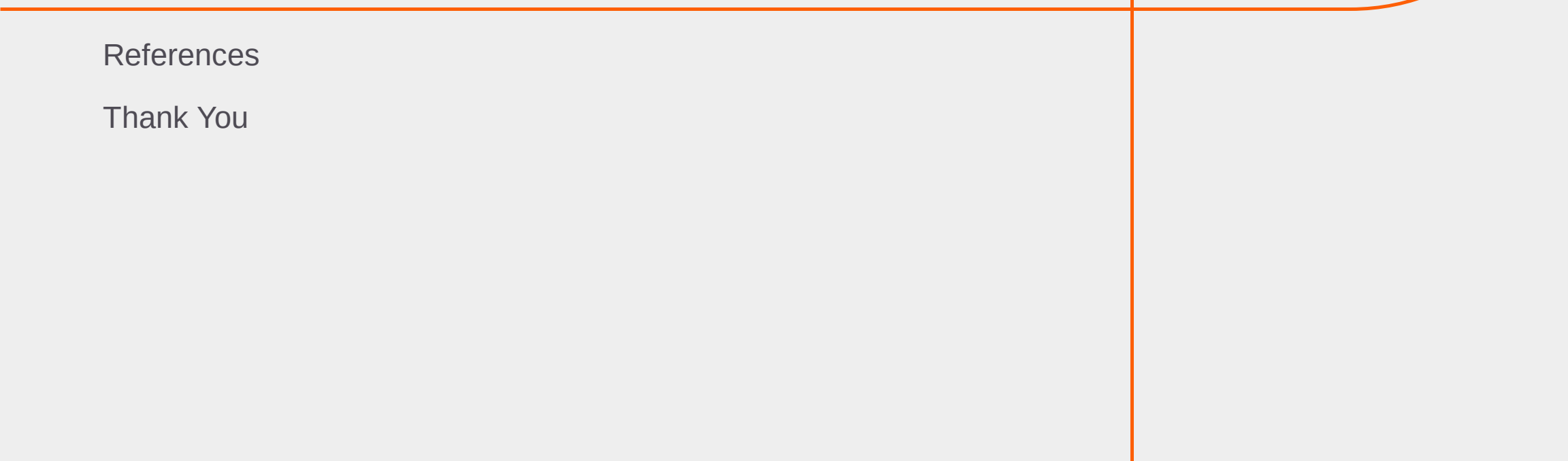
- The Angular bundle size is reduced by 40%.
- Generate and check the bundle size by running below command.
- **ng build --prod**
- For a newly created app, bundle size is pretty similar to that of Angular 8, but as your app grows, you'll see the bundle size becoming smaller vs. previous versions.

Summary: Session

With this we have come to an end of our session, where we discussed about

- Angular CLI installation
- How to create a new Angular Project
- Default folder structure and files
- How to run the project

Appendix



References

Thank You

Reference Links

- <https://github.com/angular/angular-cli>
- <https://stackoverflow.com/questions/44525746/global-angular-cli-version-greater-than-local-version>
- <https://www.geeksforgeeks.org/angular-cli-angular-project-setup/>
- <https://code.visualstudio.com/docs/nodejs/angular-tutorial>

Key Contacts :

Persistent University :

- Archana Ghatigar
archana_ghatigar@persistent.co.in



Thank you!

Persistent Interactive | Persistent University

